KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

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Multi-Vendor Marketplace DB

Data Engineering Project

Name: Shankhadeep Dey **Roll No:** 2329061 **Program:** Data Engineering

An end-to-end data engineering pipeline simulating a multi-vendor marketplace with real-time streaming, batch processing, automated data quality checks, and a live analytics dashboard.

1. Problem Statement

Modern e-commerce platforms like Amazon and Flipkart operate as multi-vendor marketplaces where thousands of vendors sell products to millions of customers simultaneously. Managing this volume of data requires robust data engineering pipelines handling both batch data and real-time streaming events. Traditional monolithic databases fail under high concurrency and cannot process real-time events efficiently — necessitating a scalable pipeline that supports batch ingestion, streaming, analytical processing, and data quality validation all working together.

2. Solution / Features

A complete end-to-end data engineering pipeline built for a Multi-Vendor Marketplace covering data generation, ETL processing, real-time streaming via Apache Kafka, batch analytics via PySpark, data quality checks, and a live Streamlit dashboard — backed by PostgreSQL with a Star Schema design.

Feature	Description
Batch Ingestion	10,000+ records generated with Faker and loaded via ETL pipeline
Real-Time Streaming	Live order events streamed through Apache Kafka every 2 seconds
Spark Batch Analytics	5 business analyses using PySpark with caching & repartitioning
Spark Structured Stream	Reads live Kafka events and processes them in real time
Data Quality	5 automated validation checks with PASS/FAIL report
Star Schema DB	Fact and dimension tables optimised for analytics
Live Dashboard	Interactive Streamlit dashboard with Plotly visualizations
Linux Automation	Bash script: directory setup, chmod 755, file movement, logging

3. Screenshots

Dashboard Overview — KPI Cards & Filters

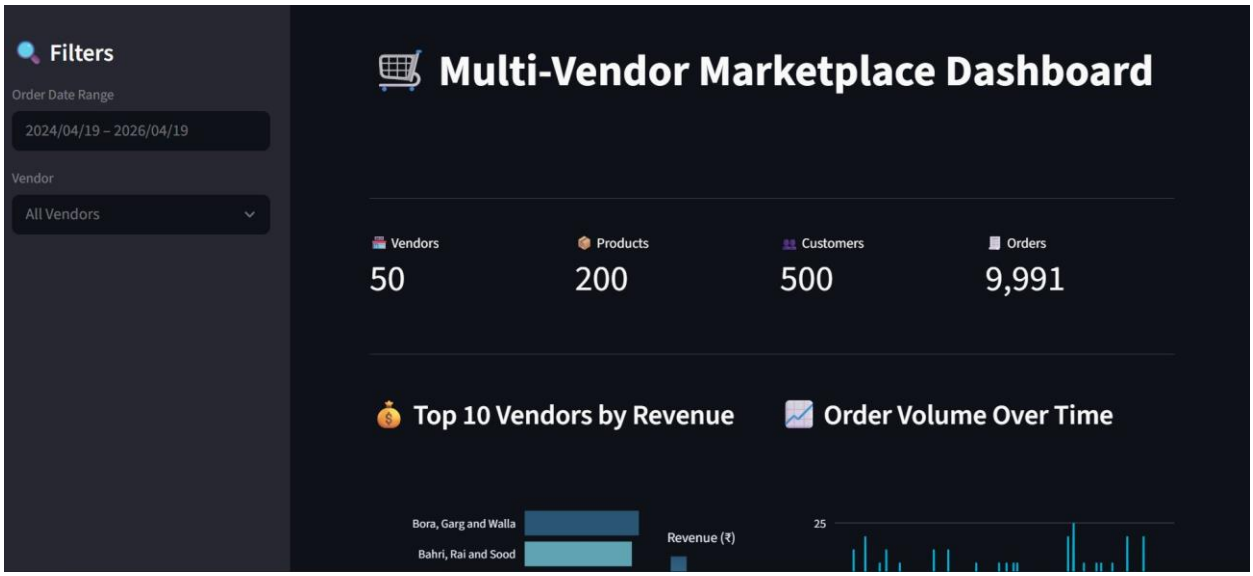


Fig 1: Live dashboard showing 50 Vendors, 200 Products, 500 Customers, 9,991 Orders with date and vendor filters

Top 10 Vendors by Revenue & Order Volume Over Time

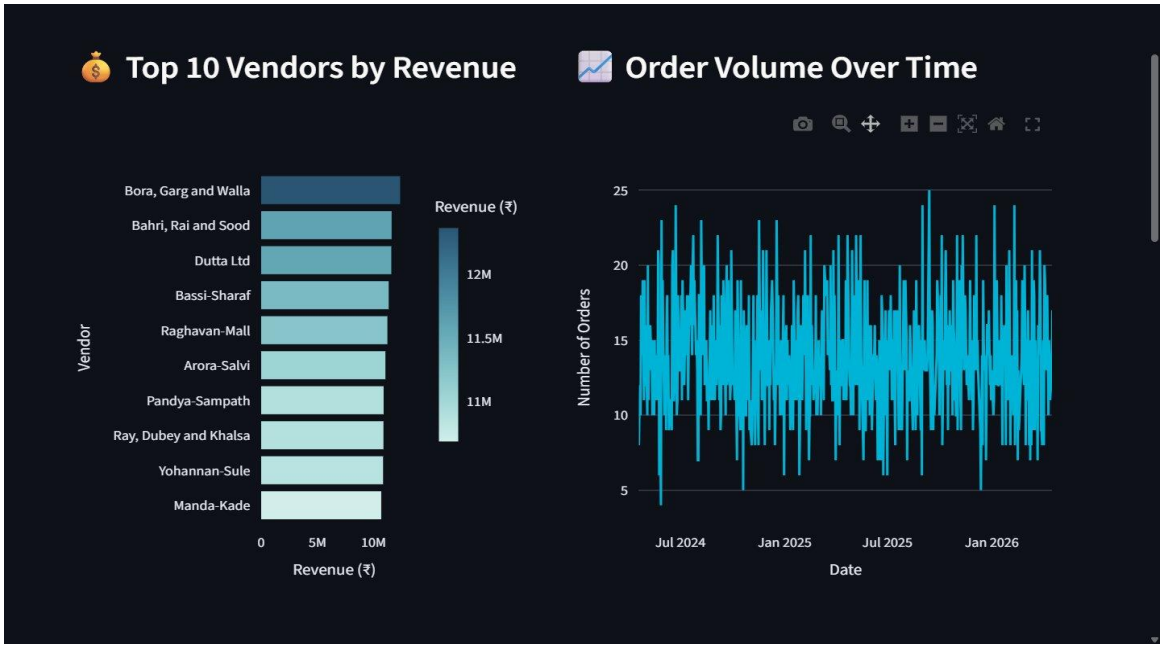


Fig 2: Bar chart — Bora, Garg and Walla leads in revenue. Line chart shows order trends from Jul 2024 to Jan 2026

Payment Status Breakdown & Products by Category

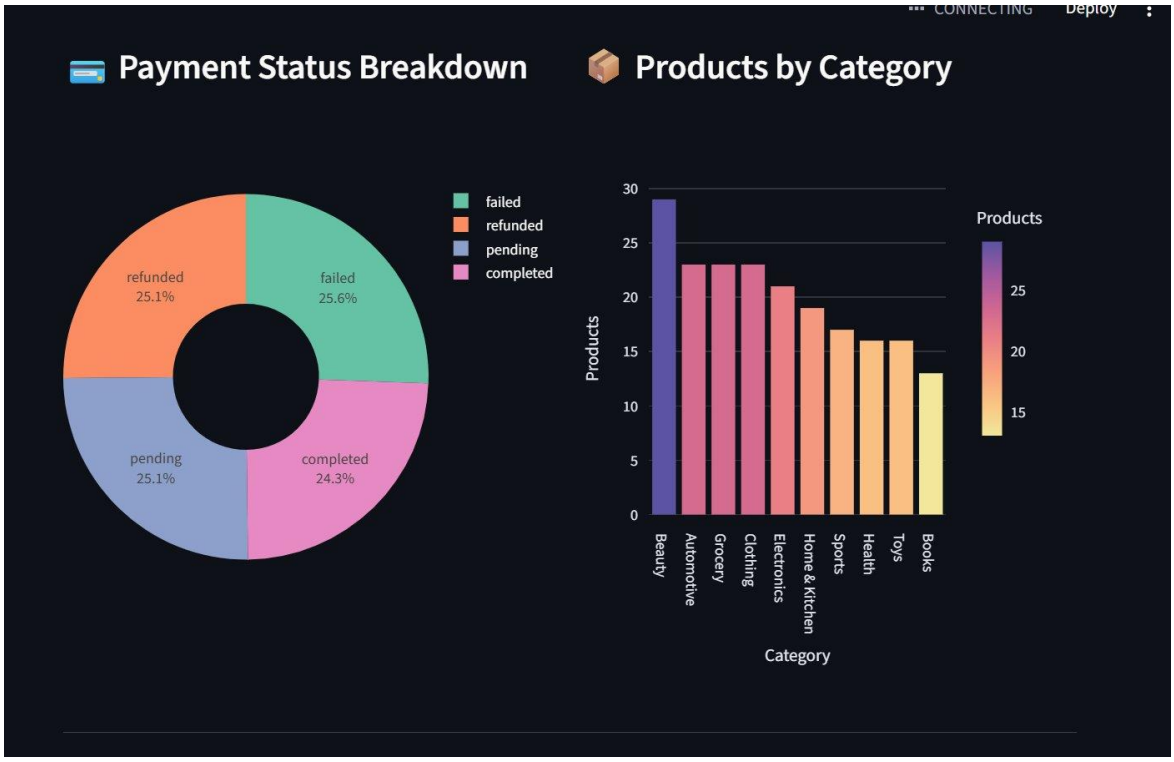
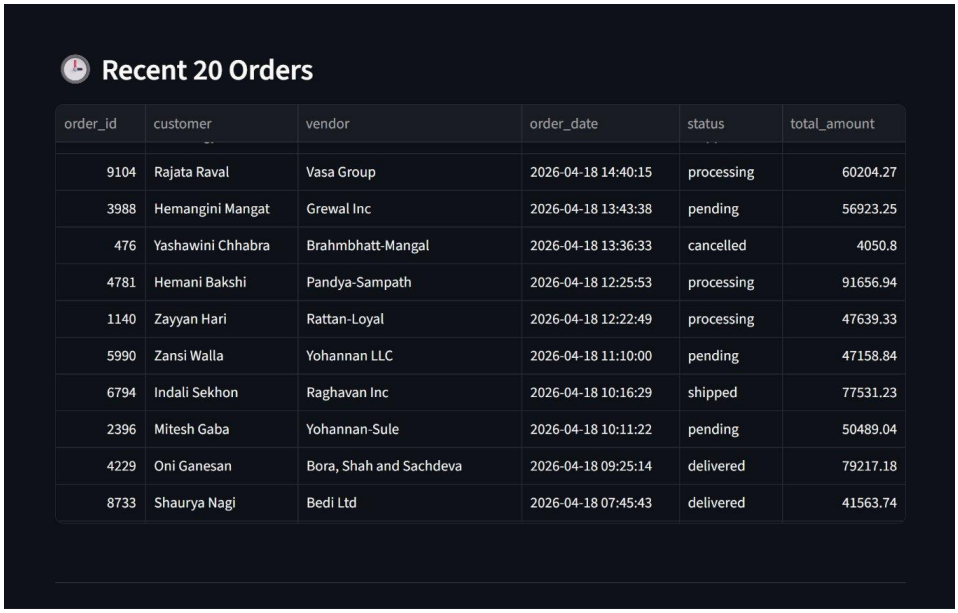


Fig 3: Donut chart shows payment split — Failed 25.6%, Refunded 25.1%, Pending 25.1%, Completed 24.3%. Category bar chart shows Beauty leads with 29 products

Recent 20 Orders Table



Recent 20 Orders

order_id	customer	vendor	order_date	status	total_amount
9104	Rajata Raval	Vasa Group	2026-04-18 14:40:15	processing	60204.27
3988	Hemangini Mangat	Grewal Inc	2026-04-18 13:43:38	pending	56923.25
476	Yashawini Chhabra	Brahmbhatt-Mangal	2026-04-18 13:36:33	cancelled	4050.8
4781	Hemani Bakshi	Pandya-Sampath	2026-04-18 12:25:53	processing	91656.94
1140	Zayyan Hari	Rattan-Loyal	2026-04-18 12:22:49	processing	47639.33
5990	Zansi Walla	Yohannan LLC	2026-04-18 11:10:00	pending	47158.84
6794	Indali Sekhon	Raghavan Inc	2026-04-18 10:16:29	shipped	77531.23
2396	Mitesh Gaba	Yohannan-Sule	2026-04-18 10:11:22	pending	50489.04
4229	Oni Ganesan	Bora, Shah and Sachdeva	2026-04-18 09:25:14	delivered	79217.18
8733	Shaurya Nagi	Bedi Ltd	2026-04-18 07:45:43	delivered	41563.74

Fig 4: Live orders table showing customer name, vendor, order date, status and amount — fed by PostgreSQL

4. Tech Stack

Layer	Technology	Version	Purpose
Language	Python	3.13	Core programming language
Data Generation	Faker	Latest	Realistic fake data generation
Data Processing	Pandas + NumPy	Latest	ETL pipeline and transformation
Database	PostgreSQL	18	Primary relational database
Streaming	Apache Kafka	4.2	Real-time event streaming (Doc
Batch Processing	PySpark	3.5	Distributed batch analytics
Stream Processing	Spark Structured	3.5	Streaming from Kafka topic
Automation	Bash Shell	—	Linux file system automation
Data Quality	Pandas (custom)	—	Validation and outlier detection
Dashboard	Streamlit + Plotly	Latest	Live interactive visualization
Containerization	Docker	Latest	Kafka deployment
JDBC Driver	PostgreSQL JDBC	42.7.10	Spark to PostgreSQL bridge

5. Unique Points

- Dual ingestion — batch (CSV to PostgreSQL via Script 3) and real-time (Kafka to PostgreSQL via Script 5) both writing to the same database simultaneously.
- PySpark optimization with cache() and repartition(4) demonstrating production-grade performance tuning on 10,000+ records.
- Both batch and stream processing implemented — Script 6 reads historical data from PostgreSQL while Script 9 reads live events from Kafka.
- Star Schema design with daily_sales_summary fact table surrounded by vendor, product, customer, and date dimensions.
- 5-check automated Data Quality framework with 3-sigma statistical outlier detection — all 5 checks passed.
- Linux bash script implementing chmod 755 file permissions, structured pipeline directories, and timestamped operation logging.
- Kafka topic partitioned into 3 partitions with consumer group offset management ensuring no event is lost or duplicated.
- Interactive Streamlit dashboard with live date range and vendor filters backed by direct PostgreSQL queries.

6. Future Improvements

- Apache Airflow DAG to orchestrate and schedule the entire pipeline automatically on a daily basis with retry logic and failure email alerts.
- Deploy on AWS EC2 with data stored in Amazon S3 as a data lake, and use Redshift or Google BigQuery for cloud-scale analytics.
- Implement Delta Lake for data versioning and time-travel queries — roll back to any previous version of the orders dataset.